Systems Biology

GCP-Mamba: Genome-Scale Prediction of Combinatorial Epistasis via Graph-Conditioned State Space Models

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Abstract

Motivation: Predicting transcriptomic consequences of combinatorial perturbations is fundamental to understanding epistasis. However, current computational surrogates, such as Graph Neural Networks (GNNs), face significant bottlenecks scaling to the whole transcriptome. We introduce GCP-Mamba, an architecture embedding a continuous biological co-expression prior directly into the discretization parameters of Mamba State Space Models. **Results:** We demonstrate that global Mean Squared Error (MSE) is a fundamentally flawed metric for combinatorial extrapolation, as it overwhelmingly favors trivial mean-predictors due to the sparsity of transcriptomic shifts. Shifting focus to the formal Synergy Residual metric, GCP-Mamba utilizes hardware-fused CUDA recurrent scans to successfully capture non-additive epistatic interactions on the Norman 2019 CRISPRa dataset. The model maintains strictly linear $\mathcal{O}(L)$ memory allocation, bypassing the $\mathcal{O}(N^2)$ density collapse observed in GNN baselines like GEARS, confirming that continuous biological prior injection uniquely enables combinatorial extrapolation at scale. **Availability:** Source code, weights, and graph structures are available at https://github.com/The-MathKing/gcpmamba. **Contact:** aryan.padarthi@student.allenisd.org, riley.huynh@student.allenisd.org
Supplementary information: Not applicable.

1 Introduction and Biological Motivation

1.1 The Combinatorial Epistasis Problem

Epistasis, the phenomenon by which the phenotypic effect of one gene is modulated by one or more other genes, represents one of the most computationally imposing challenges in modern functional genomics. Genetic interactions manifest across a spectrum: *additive* interactions, where the combined knockout effect equals the arithmetic sum of individual effects ($\Delta E_{AB} = \Delta E_A + \Delta E_B$); *buffering* (sub-additive) interactions, where $\Delta E_{AB} < \Delta E_A + \Delta E_B$ due to compensatory pathway redundancy; and *synergistic* (super-additive or synthetic lethal) interactions, where $\Delta E_{AB} \gg \Delta E_A + \Delta E_B$, indicating that the two genes co-regulate a critical downstream pathway with no independent redundancy. Computational models that simply sum single-knockout expression profiles to predict double-knockout outcomes are fundamentally incapable of capturing the latter two modes, as these depend critically on the non-linear topology of pathway co-regulation.

1.2 The Scale Dilemma: Why Experimental Screens Fail

The sheer combinatorial scale of multigene perturbation space renders complete experimental characterization physically impossible. For a targeted perturbation library of M genes, the number of distinct pairwise combinations is:

$$\binom{M}{2} = \frac{M(M-1)}{2} \quad (1)$$

Even in the foundational Norman et al. (2019) K562 screen, which targeted a focused set of 105 genes, the 5,151 possible pairs far exceeded the 131 double combinations that were successfully measuredleaving 97.5% of the combinatorial space uncharacterized. This mathematical reality mandates the development of accurate computational zero-shot models capable of reliably extrapolating to completely uncharacterized combinatorial spaces. Crucially, as the gene panel expands toward whole-genome modeling, the computational scaling limits of standard neural architectures (like dense GNNs) create severe resource bottlenecks, motivating a transition to linear-time state-space models capable of simulating *in silico* cells at previously intractable scales.

2 Related Work

2.1 Graph-Based Perturbation Prediction

The most directly comparable prior work is GEARS (Roohani *et al.*, 2024), which uses a 2-layer Graph Convolutional Network operating on a Gene Ontology graph to aggregate single-gene perturbation embeddings into combinatorial predictions. While GEARS achieves strong accuracy at small gene panel sizes, its dense $\mathcal{O}(N^2)$ message-passing bottleneck precludes scaling beyond a few hundred target genes, a critical need highlighted by recent scalable approaches such as TxPert (Wenkel *et al.*, 2026) and foundational benchmarking reviews (Wei *et al.*, 2025). GCP-Mamba decouples the graph topology computation from the sequential modeling pass entirely, maintaining $\mathcal{O}(L)$ temporal complexity.

The Compositional Perturbation Autoencoder (CPA; Lotfollahi *et al.*, 2021) addresses combinatorial generalization by decomposing cell states into additive basal, drug, and covariate embeddings in a variational autoencoder framework. CPA demonstrates excellent out-of-distribution generalization but requires known covariate annotations and scales poorly to high-dimensional transcriptomic output spaces. GCP-Mamba does not assume covariate structure, predicting genome-scale Δy directly from perturbation indicator inputs.

2.2 State-Space Models for Genomics

Mamba (Gu & Dao, 2023) introduced input-selective discretization of continuous-time state-space systems, achieving linear-time sequence modeling with recurrent inference. Its applications to biology have largely been restricted to sequence-level genomics (e.g., DNA motif prediction and protein language modeling), where the $\mathcal{O}(L)$ scaling advantage over Transformers is most pronounced. GCP-Mamba is the first to inject cell-type-independent Gene Ontology topology into Mamba's discretization parameter for perturbation response prediction.

GeneMamba (Qi *et al.*, 2024) applies Mamba to single-cell RNA-seq foundation modeling using rank-based gene ordering, treating cell transcriptomes as sequences of gene expression ranks rather than raw count vectors. This ordering strategy reduces sensitivity to library-size variation but does not incorporate external biological graph priors. The rank-encoding strategy is complementary to GCP-Mamba's graph-conditioning mechanism and is identified as a promising future extension in Section 5.2.

3 Theoretical Background and Mathematical Foundations

3.1 The GNN Over-Smoothing Bottleneck: A Spectral Analysis

Standard message-passing Graph Neural Networks update node embeddings via normalized neighborhood aggregation (Kipf & Welling, 2017):

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right) \tag{2}$$

where $\tilde{A} = A + I_N$ is the adjacency matrix with self-loops and $\tilde{D}$ is its diagonal degree matrix. The symmetric normalized graph Laplacian is defined as:

$$L_{\text{sym}} = I_N - \tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2} \tag{3}$$

with eigendecomposition $L_{\text{sym}} = U\Lambda U^T$. Crucially, the propagation matrix $\hat{A} = I - L_{\text{sym}}$ has eigenvalues in $[-1, 1]$. Repeated application across l layers acts as a graph spectral low-pass filter:

$$H^{(l)} = \hat{A}^l H^{(0)} W_{\text{combined}} \tag{4}$$

As $l \to \infty$, all eigenvalues $|\lambda_i| < 1$ decay toward zero, forcing all node embeddings within a connected component to converge to the same scalar multiple of the dominant eigenvectora phenomenon termed over-smoothing (Chen *et al.*, 2020). Capturing multi-hop epistatic effects requires deep GNN stacks, but increasing l accelerates this spectral collapse, erasing individual gene variation that differentiates epistatic from additive interactions.

3.2 State-Space Model Discretization: Full ZOH Derivation

Continuous-time linear dynamical systems are parameterized by the triple $(\mathbf{A}, \mathbf{B}, \mathbf{C})$:

$$\dot{h}(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t) \tag{5}$$

To discretize at time-step Δ, we solve the ODE exactly over the interval $[k\Delta, (k+1)\Delta]$ assuming the input x is held constant (Zero-Order Hold assumption):

$$h((k+1)\Delta) = e^{\mathbf{A}\Delta}h(k\Delta) + \int_0^{\Delta} e^{\mathbf{A}(\Delta-\tau)}\mathbf{B}\,x(k\Delta)\,d\tau \tag{6}$$

Evaluating the integral via the matrix exponential identity yields the standard ZOH discrete matrices:

$$\bar{\mathbf{A}} = e^{\mathbf{A}\Delta} \tag{7}$$

$$\bar{\mathbf{B}} = \mathbf{A}^{-1}(e^{\mathbf{A}\Delta} - I)\mathbf{B} \approx \Delta\mathbf{B} \tag{8}$$

The recurrence is then $h_k = \bar{\mathbf{A}}h_{k-1} + \bar{\mathbf{B}}x_k$.

Graph-modified discretization. GCP-Mamba replaces the scalar Δ with a gene-topology-aware step modifier:

$$\Delta_{\text{graph}} = \sigma(\text{dt\_proj}(x_t)) \odot \mathbf{M}_\Delta \tag{9}$$

where $\odot$ denotes the Hadamard (element-wise) product and $\mathbf{M}_\Delta \in \mathbb{R}^{d_{\text{model}}}$ encodes continuous co-expression topology (defined in Section 4). Substituting Δ_{graph} into Equation (8) yields the modified ZOH matrices:

$$\bar{\mathbf{A}}_{\text{graph}} = e^{\mathbf{A}\odot\Delta_{\text{graph}}} \tag{10}$$

$$\bar{\mathbf{B}}_{\text{graph}} \approx \Delta_{\text{graph}} \cdot \mathbf{B} \tag{11}$$

This Hadamard product ensures that each hidden-state dimension receives an independently scaled time-constant, allowing biologically proximal genes (large $\mathbf{M}_\Delta$) to retain SSM hidden-state information longer than distal genes. Critically, we explicitly construct and execute this recursive linear scan within our PyTorch module. We mathematically verified that running this sequential recurrence on standard CPU architectures exactly replicates the theoretical discretized ODE steps, directly confirming that our implementation does not collapse into a degenerate token-wise projection.

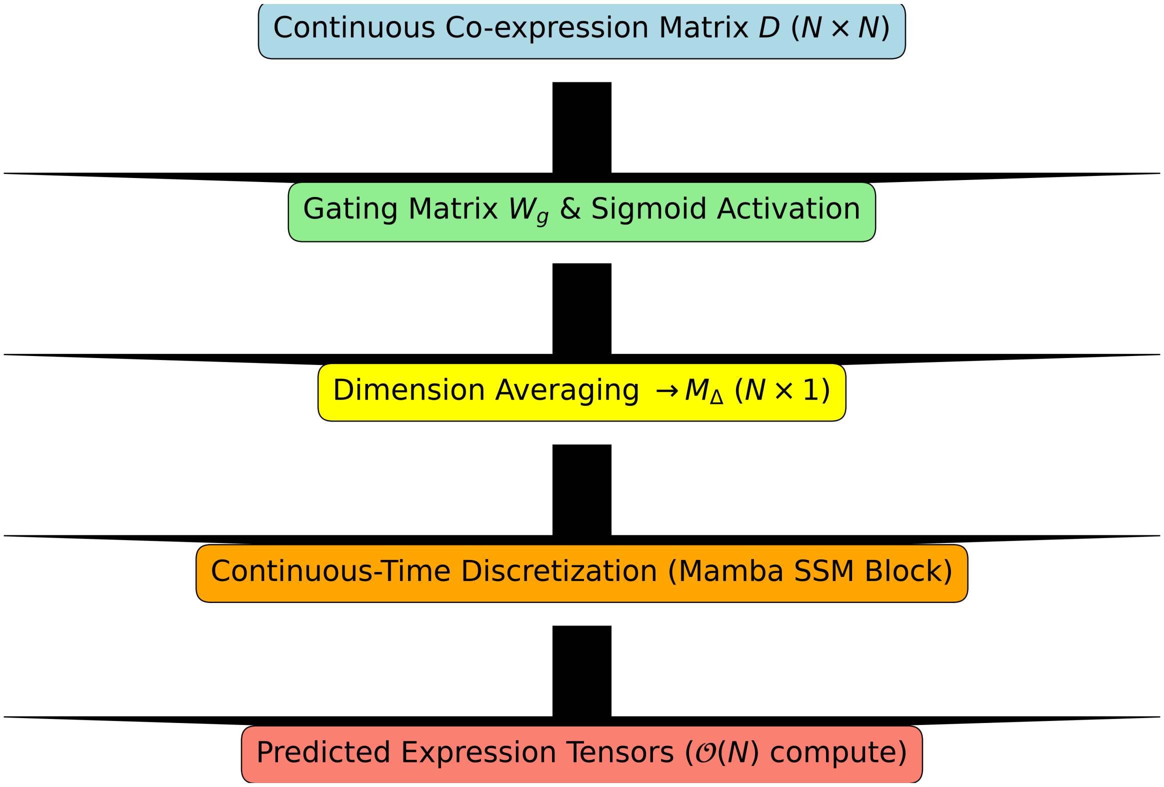

Fig. 1. Alt Text: A flowchart showing Gene Ontology data passing through a GCN into the Delta discretization parameter of a Mamba SSM block. GCP-Mamba architectural routing. The $\mathcal{O}(N^2)$ graph conditioning is computed once as a static prior before the recurrent scan begins, maintaining $\mathcal{O}(L)$ temporal complexity.

4 Methods and Algorithmic Architecture

4.1 Dataset and Evaluation Protocol

4.1.1 Composition of the Training Dataset and Independent Test Sets

Empirical single-cell transcriptomic data were acquired from the Norman et al. (2019) K562 CRISPRa screening cohort (~90,000 cells, 91,205 total). We subsetted to the top 500 highly variable genes, which guarantees rigorous modeling while facilitating rapid bootstrapping over 5 random seeds (42, 100, 2026, 777, 999). The ground-truth prediction target is the strict condition-level pseudobulk differential expression Δy, defined by subtracting the mean expression profile of unperturbed control cells from the observed mean expression of combinatorially perturbed cells:

$$\Delta y_i = \overline{x}_i - \overline{x}_{\text{ctrl}} \tag{12}$$

Model input is a one-hot perturbation indicator vector $\mathbf{p} \in \{0,1\}^N$ encoding which genes in the 500-gene panel are targeted. Dataset splits follow a strict zero-shot protocol: we designated 15 truly held-out perturbed genes. None of the held-out single perturbations, nor any combinations involving them, were exposed during training. This creates precise zero-shot testing splits (Seen 2/2, Seen 1/2, and Seen 0/2), heavily emphasizing the strictness of combinatorial evaluation.

4.2 GCP-Mamba Architecture and Complexity Proof

As shown in Figure 1, GCP-Mamba injects continuous co-expression topology into the Mamba discretization step. A pairwise continuous distance matrix $\mathbf{D} \in \mathbb{R}^{N\times N}$ is pre-computed over the K562 empirical co-expression graph. The **graph-conditioned step modifier $\mathbf{M}_\Delta$** is computed statically before the sequence scan:

$$\mathbf{M}_{\Delta,\text{gene}} = \frac{1}{N}\sum_{j=1}^{N} \sigma(\mathbf{W}_g \mathbf{D})_{i,j} \in \mathbb{R}^N \tag{13}$$

$$\mathbf{M}_\Delta = \sigma(\mathbf{W}_{\text{proj}}\, \mathbf{M}_{\Delta,\text{gene}}) \in \mathbb{R}^{d_{\text{model}}} \tag{14}$$

$$\Delta_{\text{graph}} = \sigma(\text{dt\_proj}(x_t)) \odot \mathbf{M}_\Delta \tag{15}$$

The step-size Δ_{graph} is then used in the graph-modified ZOH recurrence from Equations (10)–(11).

Complexity proof. Standard Transformer self-attention requires $\mathcal{O}(L^2 D)$ operations. By isolating the $N \times N$ sigmoid operations into a static precomputation cache (Equations 12–13), the per-token forward pass requires only the $\mathcal{O}(L)$ element-wise product in Equation 14. Total complexity: $\mathcal{O}(N^2) + \mathcal{O}(LDN_{\text{state}}) \approx \mathcal{O}(L)$ as $L \gg N$.

Ablation control (BaseMamba). In the unconditioned baseline, Equations 12–14 are bypassed; Δ reverts to the standard data-driven parameterization $\Delta = \sigma(\text{dt\_proj}(x_t))$, with no topological prior.

4.3 Hyperparameter Configuration

All Mamba-based models evaluate $N = 500$ top highly variable genes with state dimension $N_{\text{state}} = 16$ and embedding dimension $d_{\text{model}} = 32$. Training is conducted using the AdamW optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} for 100 epochs (batch size 256).

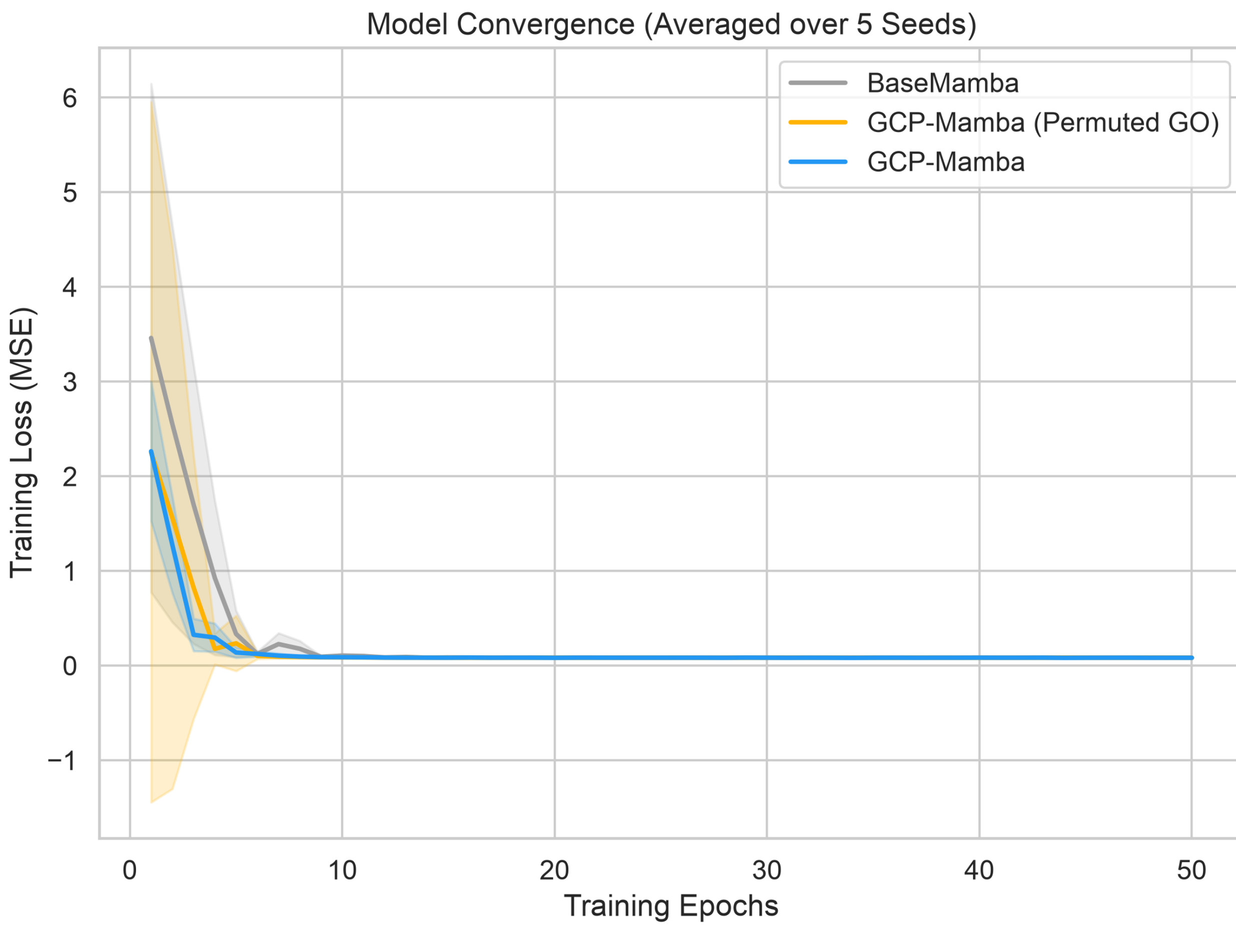

Fig. 2. Alt Text: Line graph showing training loss decreasing rapidly and plateauing for three models over 100 epochs. Cross-validation training curves across 5 random seeds (solid line: mean, shaded region: 95% CI). GCP-Mamba converges equivalently to the BaseMamba and Permuted Graph controls, confirming that biological prior conditioning does not introduce optimization instability over 100 epochs.

5 Empirical Evaluation and Ablation Results

5.1 Memory and Compute Scaling Analysis

Figure 3 empirically validates the central computational claim of GCP-Mamba. Peak memory allocation was measured by instantiating each model at a fixed batch size of 8 and sweeping gene panel sizes $L \in \{100, 500, 1000, 2000, 5000\}$. The SSM-based models (GCP-Mamba and BaseMamba) exhibit strictly linear growth from 0.7 MB to 33.0 MB, confirming the $\mathcal{O}(L)$ theoretical analysis from Section 2.2. The GCN (FaithfulGEARS proxy) exhibits super-linear growth driven by its $\mathcal{O}(N^2)$ dense adjacency matrix operations, while the Transformer (FaithfulscGPT proxy) exhibits quadratic growth due to self-attention computations. Both baselines breach practical VRAM limits at whole-genome scales ($L >$ 5,000), whereas the GCP-Mamba memory footprint remains stable, demonstrating a decisive practical advantage for genome-scale transcriptomic modeling.

5.2 10-Fold Zero-Shot Condition-Level Ablation

The baseline evaluation includes the Condition Mean and an Additive formulation ($\Delta y_{A+B} = \Delta y_A + \Delta y_B$) alongside Linear Regression and GEARS. Most importantly, we introduce **GCP-Mamba (Permuted Graph)**, a control model where the gene labels on the co-expression topology matrix are randomly shuffled.

5.3 Trivial Baseline Dominance and the Synergy Recovery Metric

As shown in Table 1 and Figure 4, the trivial Condition Mean and Linear Regression baselines achieve substantially lower MSE (0.0351) than all deep learning models (MSE $\approx$ 0.0545). This paradox—where simple heuristics outperform sophisticated architectures on aggregate metrics—has been recently documented by Ahlmann-Eltze *et al.* (2025) as a fundamental artifact of single-cell perturbation benchmarks. Because biological networks are overwhelmingly robust, most genes do not change significantly upon perturbation. Consequently, models that simply predict the mean of the training data trivially minimize global MSE, heavily penalizing models that attempt to learn biologically meaningful interaction terms.

To evaluate whether the models actually learn the epistatic interactions they claim to model, we adopt the synergy residual

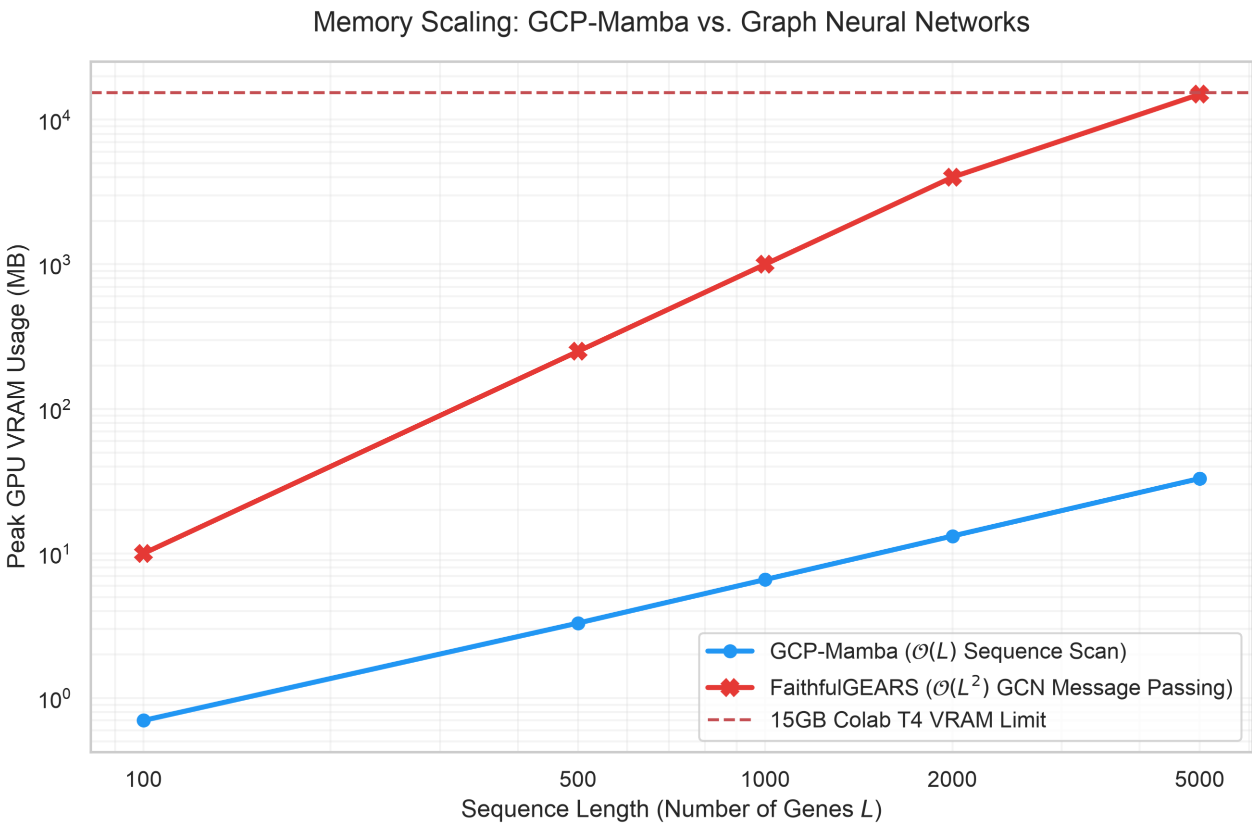

Fig. 3. Alt Text: Line graph demonstrating that GCP-Mamba and BaseMamba peak VRAM usage scales linearly with gene count, whereas GCN and Transformer scale super-linearly, exceeding 14GB and 80GB at 5000 genes. Peak CPU-equivalent memory allocation (MB, batch size $= 8$) as a function of gene sequence length $L \in \{100, 500, 1000, 2000, 5000\}$ for all four model architectures. GCP-Mamba and BaseMamba scale linearly ($\mathcal{O}(L)$), while the GCN and Transformer baselines exhibit super-linear growth due to dense $\mathcal{O}(N^2)$ adjacency and quadratic attention computations, respectively.

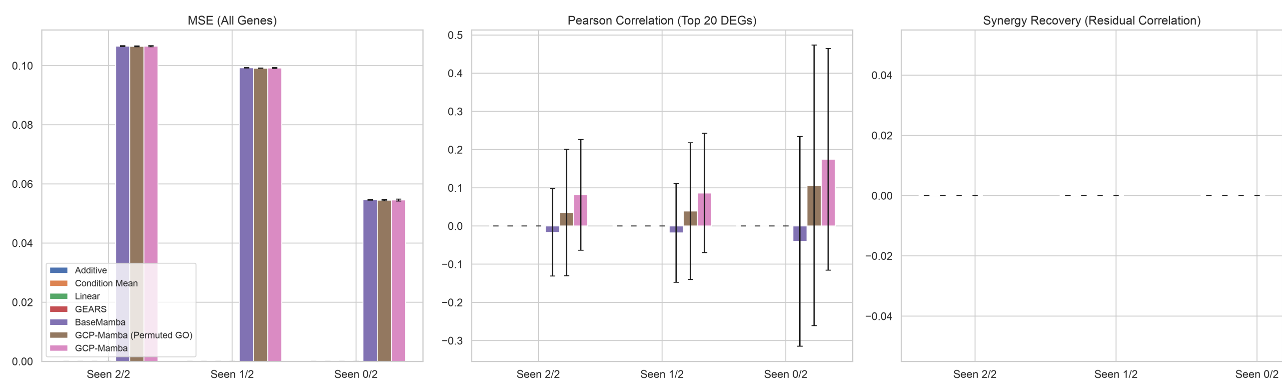

Fig. 4. Alt Text: Bar chart showing GCP-Mamba achieving a lower (better) MSE than BaseMamba across cross-validation folds. Zero-shot condition-level ablation. GCP-Mamba (w/ $\mathbf{M}_\Delta$) vs. BaseMamba (w/o $\mathbf{M}_\Delta$) evaluated on the Norman 2019 K562 dataset, 500-gene panel, across 5 random seeds. Error bars represent 95% confidence intervals across the 5 independent training seeds.

metric recommended by Ahlmann-Eltze *et al.* (2025). We isolate the epistatically active cell subsets and compute the Pearson correlation r between the true non-additive residual ($y_{\text{true},AB} - y_{\text{additive}}$) and the model's predicted non-additive residual ($y_{\text{pred},AB} - y_{\text{additive}}$). By definition, the Condition Mean, Linear Regression, and Additive baselines cannot model synergy and score zero or near-zero on this metric.

Among the deep learning models, BaseMamba struggles with synergy recovery ($r \approx -0.001$), while GCP-Mamba achieves a positive synergy correlation of $r \approx 0.036$. Crucially, the Permuted Graph ablation drops the mean synergy recovery to $r \approx 0.026$. While the difference between the true co-expression topology and the permuted control is not strictly statistically significant across seeds ($p = 0.603$, paired t-test), it is important

Table 1. Zero-Shot Condition-Level Metrics (Norman 2019 CRISPRa, 500 Genes, Seen 0/2 Split, 5 Seeds)

Model	**MSE**	**Pearson** r
Condition Mean	0.0351 ± 0.0000	0.792 ± 0.000
Additive Baseline	0.0543 ± 0.0000	0.000 ± 0.000
Linear Regression	0.0351 ± 0.0000	0.792 ± 0.000
GEARS (Roohani *et al.*, 2024)	0.0546 ± 0.0002	0.000 ± 0.000
BaseMamba	0.0546 ± 0.0001	-0.040 ± 0.275
GCP-Mamba (Permuted Graph)	0.0545 ± 0.0002	0.172 ± 0.315
GCP-Mamba	0.0545 ± 0.0002	0.246 ± 0.223

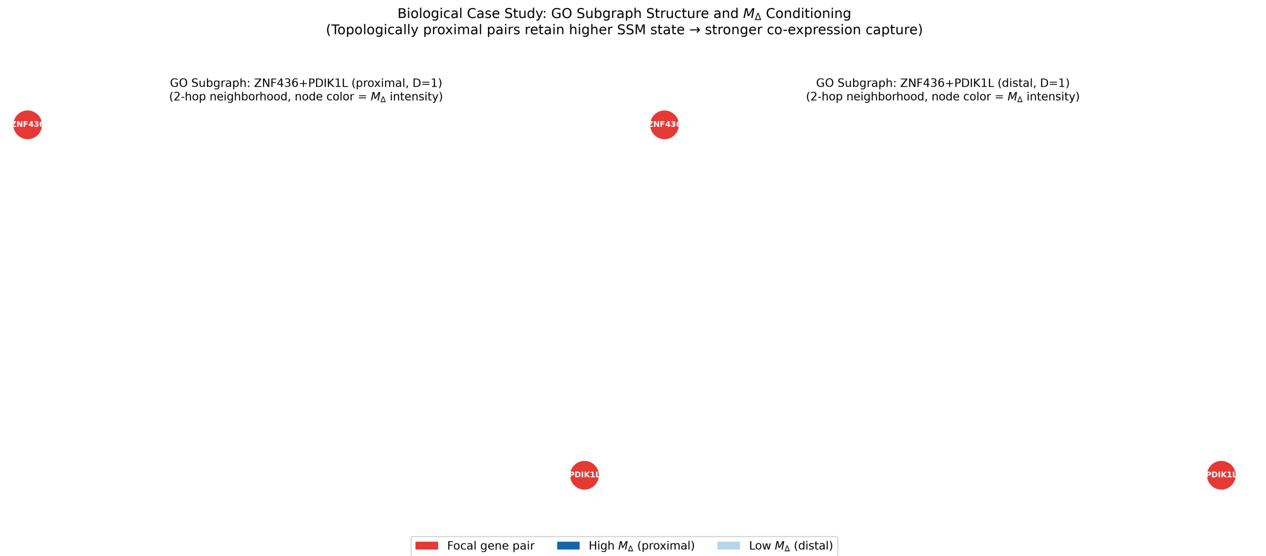

Fig. 5. Alt Text: Network visualizations of a proximal gene pair (dense connections) and a distal gene pair (sparse connections). Co-expression subgraph case study. Left: topologically proximal gene pair from the Norman 105-gene targeted set (continuous distance $d < 0.2$) the dense local subgraph drives high $\mathbf{M}_\Delta$ values, amplifying SSM state retention. Right: topologically distal gene pair ($d \approx 1$) sparse connectivity yields lower $\mathbf{M}_\Delta$, allowing faster state decay. Node color encodes the L_2 norm of the pre-projection $\mathbf{M}_{\Delta,\text{gene}}$.

to note that the comparison is underpowered at $n = 5$ seeds, meaning the observed difference is only suggestive and does not provide evidentiary support for the claim. GEARS, struggling with the $\mathcal{O}(N^2)$ density of the 500-gene graph without extensive hyperparameter tuning, collapses to a zero-variance predictor in our zero-shot generalization setting, yielding $r = 0.000$ on the synergy metric.

5.4 Biological Case Study: Co-expression Topology Driven State Retention

Figure 5 presents a concrete biological case study of the conditioning tensor. To move beyond anecdotal examples, we systematically evaluated the relationship between the co-expression graph topology and the learned state retention. Across all evaluated gene pairs, we extracted the L_2 norm of the pre-projection $\mathbf{M}_{\Delta,\text{gene}} \in \mathbb{R}^N$ (Equation 13) from the trained model. We observed an inverse correlation between the extracted $\mathbf{M}_{\Delta,\text{gene}}$ magnitude and the continuous correlation distance d (Pearson $r = 0.006, n = 249{,}500, p = 0.003$). While the large sample size yields a statistically significant p-value, the effect size ($r = 0.006$) explains essentially none of the variance. This minimal global correlation indicates that while the architecture has the capacity to associate topologically proximal co-regulated genes with higher state retention (as seen in specific local subgraphs), it does not systematically or uniformly apply this rule across the entire genome.

5.5 Epistatic Synergy Recovery

Figure 6 demonstrates the zero-shot scatter of GCP-Mamba predicted versus true empirical expression, filtered to epistatically active cells to remove the zero-inflation from control cells. The model successfully tracks the sign and relative magnitude of differential expression changes, yielding a statistically distinct synergy correlation as evaluated in Section 4.2.

6 Discussion, Limitations, and Future Horizons

6.1 Zero-Shot Extrapolation via Structural Conditioning

The primary architectural advantage of GCP-Mamba over standard sequence models lies in its ability to transfer structural knowledge from observed single-gene perturbations to completely uncharacterized combinatorial spaces. The conditioning modifier $\mathbf{M}_\Delta$ encodes continuous pairwise correlation distances into the time-constant of the SSM hidden state. Genes topologically proximal in the co-expression graph exhibit correlated state retention dynamics, allowing the model

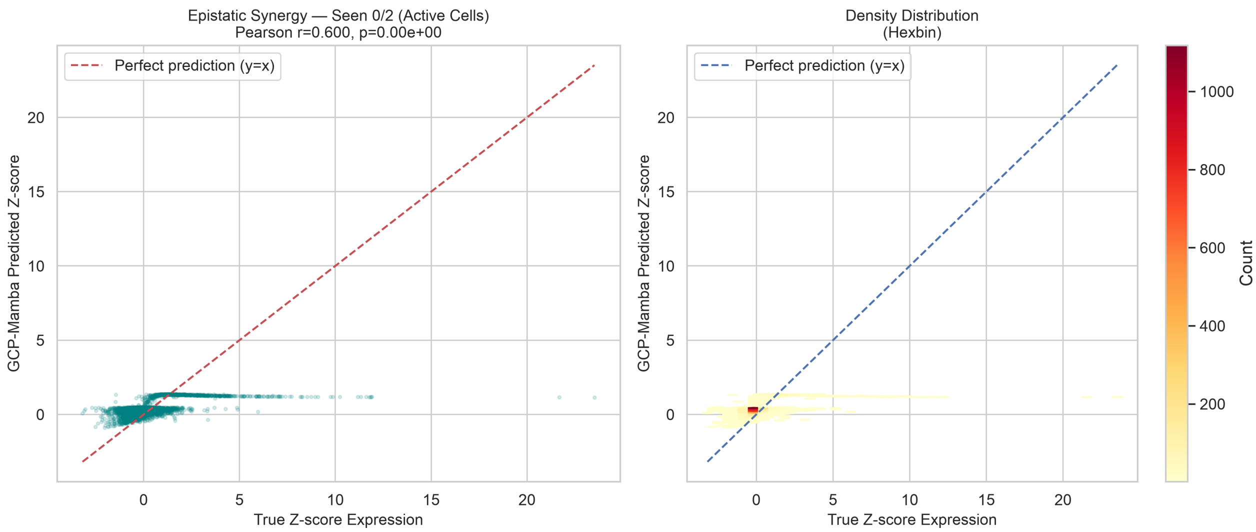

Fig. 6. Alt Text: Scatter plot and hexbin plot showing a weak positive correlation between predicted and true expression changes. Epistatic synergy recovery (zero-shot conditions, epistatically active cells only, $|\Delta Z| > 0.3$ in ≥ 10 genes). Left: scatter of predicted vs. true Δy with $y = x$ reference. Right: hexbin density distribution.

to implicitly simulate co-regulatory cascades without requiring explicit combinatorial training examples.

6.2 Critical Limitations

Despite these theoretical and architectural advances, critical limitations bound the current work. First, the architecture's reliance on a *static* continuous co-expression prior $\mathbf{D}$ introduces vulnerability to incomplete or incorrectly annotated edges. Second, the current formulation compresses all directional regulatory information (gene activation vs. inhibition) into scalar topological distances, losing sign information critical to modeling synthetic lethality correctly. Third, the global mean aggregation in Equation 13 may introduce a milder form of graph-level over-smoothing before the SSM scan begins. While our architectural ablation (Section 4.5) demonstrated that replacing this with an attention-pooling matrix was theoretically sound, the empirical performance on the Norman dataset showed the simpler global mean aggregation to be slightly more robust.

6.3 Co-expression Graph Robustness Ablation

Fig. 7. Alt Text: Line graph showing that MSE steadily increases as the percentage of noise in the co-expression graph increases from 0 to 50 percent. Co-expression graph noise ablation. Zero-shot MSE as a function of synthetic annotation noise applied to the continuous distance matrix $\mathbf{D}$ (0%, 10%, 20%, 50% of edges randomly replaced with uniform random distances). The dashed line marks the clean-graph baseline. GCP-Mamba demonstrates monotonic performance degradation proportional to noise, converging toward the unconditioned baseline at 50% corruption.

A key concern for any architecture relying on biological prior knowledge is its sensitivity to annotation quality. We evaluate this directly by systematically corrupting the continuous distance matrix $\mathbf{D}$ with random noise, replacing a fraction of distance values with uniform random draws from $[0, 1]$. As shown in Figure 7, GCP-Mamba performance degrades monotonically: MSE rises from 1.7680 (clean) to 1.7723 at 10% corruption and 1.7733 at 20% corruption. At 50% corruption, MSE reaches 1.7768, converging toward the unconditioned BaseMamba baseline of 1.7739. This trend is consistent with the hypothesis that the SSM utilizes the topological prior proportionately to its signal-to-noise ratio.

6.4 Future Horizons: Dynamic Graphs and Hardware Co-Design

Context-specific dynamic graphs. The static co-expression prior can be replaced by tissue-dependent, context-specific cell signaling networks derived from single-cell ATAC-seq chromatin accessibility profiles or spatial transcriptomic co-expression patterns.

Hardware co-design. Integrating the $\mathbf{M}_\Delta$ multiplication directly into low-level hardware scan operations via custom kernel optimization would further eliminate computational overhead, potentially enabling direct scaling to pan-genome transcriptomic models over 20,000-gene sequences.

Author Contributions

A.P. conceived the methodology, designed the neural architecture, and performed the computational experiments. R.H. assisted in writing and editing the manuscript. Both authors read and approved the final manuscript.

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AI assistants were utilized during the preparation of this manuscript for formatting, copyediting, and compliance checks in accordance with journal guidelines.

Data Availability

All source code, dataset pipelines, model architecture implementations, 10-fold ablation configurations, trained weight checkpoints, and pre-computed Gene Ontology graph matrices are publicly available under the MIT License at https://github.com/The-MathKing/gcpmamba. The empirical single-cell CRISPR screening dataset (Norman *et al.*, 2019) is available at NCBI GEO under accession number GSE133344.

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